

Niranjan C N

☎ +91 9108269436

✉ cnniranjan72@gmail.com

🌐 linkedin.com/in/niranjan-c-n

🐙 github.com/cnniranjan72

Summary

Software Engineer specializing in AI systems and backend engineering with experience building production-grade AI platforms, scalable backend services, RAG pipelines, and enterprise knowledge systems using Python, FastAPI, PostgreSQL, Redis, Docker, and modern LLM technologies.

Education

AMC Engineering College — B.E. Computer Science & Engineering (CGPA: 8.5)

2023 – 2027

Experience

AI Developer, Tribeca Softech — Remote

Jan 2026 – Present

- Architected production-grade AI backend services using **FastAPI**, **PostgreSQL (Neon)**, and asynchronous processing for scalable enterprise knowledge systems.
- Designed distributed ingestion and vectorization pipelines using **AWS S3**, **Redis**, **BullMQ**, and **pgvector** for scalable semantic retrieval.
- Built retrieval, evaluation, telemetry, and observability pipelines for enterprise RAG systems.
- Implemented GraphQL APIs, JWT authentication, RBAC, structured logging, Docker deployment, health monitoring, and workflow orchestration.
- Led architecture and implementation of multiple AI platform components emphasizing scalability, reliability, maintainability, and production readiness.

Software Development Engineer, AeroAspire — Hybrid

Sep 2025 – Present

- Developed **AeroBriefs**, a cross-platform aviation weather briefing application using Flutter, Firebase Authentication, Firestore, and REST APIs.
- Built reliable synchronization pipelines for real-time aviation operations following industry best practices.

Projects

BASTION — AI Agent Governance & Runtime Control Plane

github.com/cnniranjan72/BASTION

- Built and deployed a production AI-agent governance platform (**FastAPI**, **PostgreSQL**, **Redis**, **Kafka**, **Docker**, **Kubernetes**) that intercepts and evaluates every agent tool call pre-execution, blocking, allowing, or escalating to human approval in real time; validated by **332/332 tests** against live infrastructure (no mocks).
- Designed an event-sourced architecture with a transactional outbox and Kafka-based distribution for durable, idempotent, at-least-once event delivery, feeding a live 3D execution-graph frontend (React, react-three-fiber) over Web-Sockets; implemented policy hot-reload across running instances via Redis pub/sub and RBAC across 4 roles.
- Profiled and resolved production performance bottlenecks using **py-spy** (OpenTelemetry export batching, database connection-pool sizing), and extended the platform to govern a real transacting agent (Razorpay integration) with policy-enforced spend limits and rate-based anomaly blocking; **100% prevention across 20/20** live prompt-injection attempts.

AIMirror — Behavioral AI Digital Twin Platform

github.com/cnniranjan72/AI-Mirror

- Built a production-grade behavioral digital twin platform that transforms Instagram Reels and YouTube activity into an explainable cognitive identity model using **Chrome Extension**, **FastAPI**, **React**, **PostgreSQL + pgvector**, and Docker.
- Engineered a deterministic cognitive pipeline spanning **behavioral ingestion**, **knowledge consolidation**, **5-dimensional evidence scoring**, **rule-based inference**, **9-profile identity construction**, **temporal snapshots**, and **contextual-bandit personalization**, with **384-dimensional** vector retrieval and multi-provider LLM verbalization.

Skills

Languages: Python, Java, C++, JavaScript, TypeScript, Dart, C, HTML, CSS

Frameworks: FastAPI, Flask, Node.js, Express.js, React.js, Flutter, GraphQL

AI / ML: LLMs, Agentic AI, RAG, Vector Databases, Embeddings, Semantic Search

Databases: PostgreSQL, Redis, MongoDB, MySQL, Firestore, Pinecone, pgvector

Cloud & Infrastructure: Docker, Kubernetes, Linux, AWS S3, BullMQ, Kafka

Developer Tools: Git, Postman, Swagger, Playwright

Additional Information

Certifications: Anthropic - Claude Code in Action, AI Fluency, AWS Cloud Practitioner Essentials, MongoDB for Students, The Complete Full-Stack Web Development Bootcamp

Languages: English, Kannada, Hindi, Telugu